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An Architectural Framework for 6G Network Digital Twins System

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Abstract

In the context of 6G, Digital Twin technology has the potential to play a crucial role by offering a sophisticated, dynamic model that mirrors the physical world in real-time. Integrating digital twins into 6G enables precise orchestration and optimization of network resources, meeting the diverse and stringent demands of next-generation applications. Furthermore, digital twins provide a platform for continuous learning and AI-driven analysis. Research on the architecture of Network Digital Twins (NDTs) for 5G/6G is still limited, often focusing on partial implementations rather than comprehensive, full-stack approaches. This paper proposes a high-level architectural framework for 6G NDTs system, structured across three layers: the Physical Twin Layer, Digital Twin Layer, and Application and Service Layer. We discuss the challenges of deploying these systems and the importance of selecting appropriate tools, presenting an experimental case study that demonstrates the impact of different tool choices on system performance. Our findings underscore the need for flexible, scalable solutions to fully realize the benefits of NDTs in 6G networks.

CCS Concepts

• **Networks** → **Network architectures; Network components; Network management.**

Keywords

6G, Mobile Network, Network Digital Twin, AI-driven Network

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1 Introduction

The concept of Digital Twins (DTs) has seen significant growth in recent years. The application and evaluation of DTs within network systems remain relatively underdeveloped. As the telecommunications industry gears up for 6G networks—offering unprecedented improvements in speed, capacity, and latency—the relevance of Network Digital Twins (NDTs) comes into focus.

NDTs serve as sophisticated virtual replicas of network setups and operational strategies. Research on DTs in the network has just begun, and its application is still in its infancy stage [1]. Most of the cases and studies that claim to introduce NDTs in networks, such as 5G, are partial NDTs. There is currently a lack of a comprehensive digital twins system that integrates multiple interconnected digital twins to model a more complex and unified architecture, particularly for the next-generation 6G network.

To address these gaps, we propose a high-level architectural framework for 6G Network Digital Twins system, consisting of three key layers: the Physical Twin Layer, Digital Twin Layer, and Application and Service Layer, along with potential strategies for deploying with appropriate tools.

The remainder of the paper is organized as follows. Section 2 reviews related literature and approaches. Section 3 introduces the proposed NDTs system architecture. Section 4 benchmarks existing potential software tools, providing recommendations for future tool selection. Finally, Section 5 concludes the paper.

2 Related Work

Establishing a unified and standardized structure for digital twins remains challenging [2]. Tao et al. pointed out the dimensions of general DTs [3], while Zhou et al. primarily



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explored the overall architecture and the functions that NDTs should perform [4]. However, limited consensus exists on the specific details and modular components, which restricts the development of a well-defined structure and hinders effective guidance for future practical applications.

For practical development, Ericsson implemented a 5G NDT [5], providing a virtual environment for optimizing radiated power. This design embodies NDTs' ideas, using simulation and AI tools. HEAVY.AI developed an NDT application with a three-layer framework called HeavyRF [6], capable of real-time RF simulations on high-resolution terrain data. Additionally, Demir et al. [7] optimized beam selection by extracting spatial distribution of scatterers from user-side camera images.

However, most studies focus on specific aspects, with less attention on the full network and system-wide cooperation [6]. Therefore, our work takes a higher-level approach, proposing a modular architecture and exploring principles and strategies for implementing a more comprehensive NDTs system.

3 High-level 6G NDTs Architecture

In this section, we proposed a three-layer general 6G NDTs system structure, drawing on existing architectures and implementations to offer practical guidance for deployment. The proposed architecture includes the Physical Twin Layer, Digital Twin Layer, and Application and Service Layer, as shown in Figure 1.

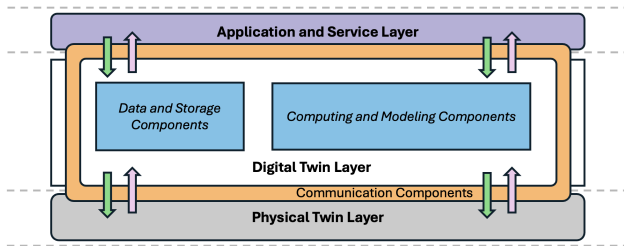


Figure 1: 6G Network Digital Twins system high-level architecture. From bottom to top: Physical Twin Layer, Digital Twin Layer, and Application and Service Layer

3.1 Physical Twin Layer

The Physical Twin Layer is primarily responsible for gathering and preliminary processing data from the physical.

In terms of composition requirements, as shown in Figure 2, objects in the physical world that can be modeled as digital objects should be included [8]. Network hardware like User Equipment (UE) and gNodeB are thus covered in our architecture. Contextual data, including location and

material properties that affect signal attenuation, must also be integrated for a more comprehensive twins system.

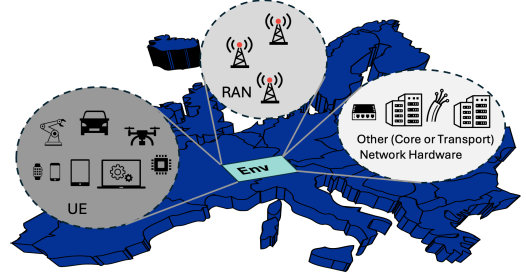


Figure 2: Physical Twin Layer, including network devices and environment

In terms of performance requirements, the Physical Twin Layer should support high-frequency, efficient, and reliable data synchronization for seamless updates and command execution between physical and digital components. 6G networks must enable ultra-fast data transmission, low latency, and extensive connectivity. Energy efficiency, security, and seamless integration are also critical aspects.

3.2 Digital Twin Layer

An illustration of the Digital Twin Layer is shown in Figure 3. It includes digital counterparts and extended capabilities. In functional components, we propose that this layer should minimally include three core components: Communication Components, Data and Storage Components, and Computing and Modeling Components.

3.2.1 Communication Components. In Figure 3, the Communication Components are highlighted by the orange sections surrounding the system. They act as the essential link between physical and virtual objects and virtual-to-virtual communication. They manage uninterrupted data transmission and transformation to manipulate the physical network. This module converts raw data into structured formats for system compatibility. Depending on specific requirements, middleware may also be involved to facilitate communication across heterogeneous systems.

3.2.2 Data and Storage Components. The Data and Storage Components manage interconnected processes: data transformation, storage, and protection. Temporary storage supports rapid processing, and metadata storage aids in efficient data organization and retrieval. Data protection measures like encryption and access control ensure safety and integrity throughout the data lifecycle. Inputs come from multiple sources, including real-time systems and simulations, with data transformed and stored for future analysis or AI model training.

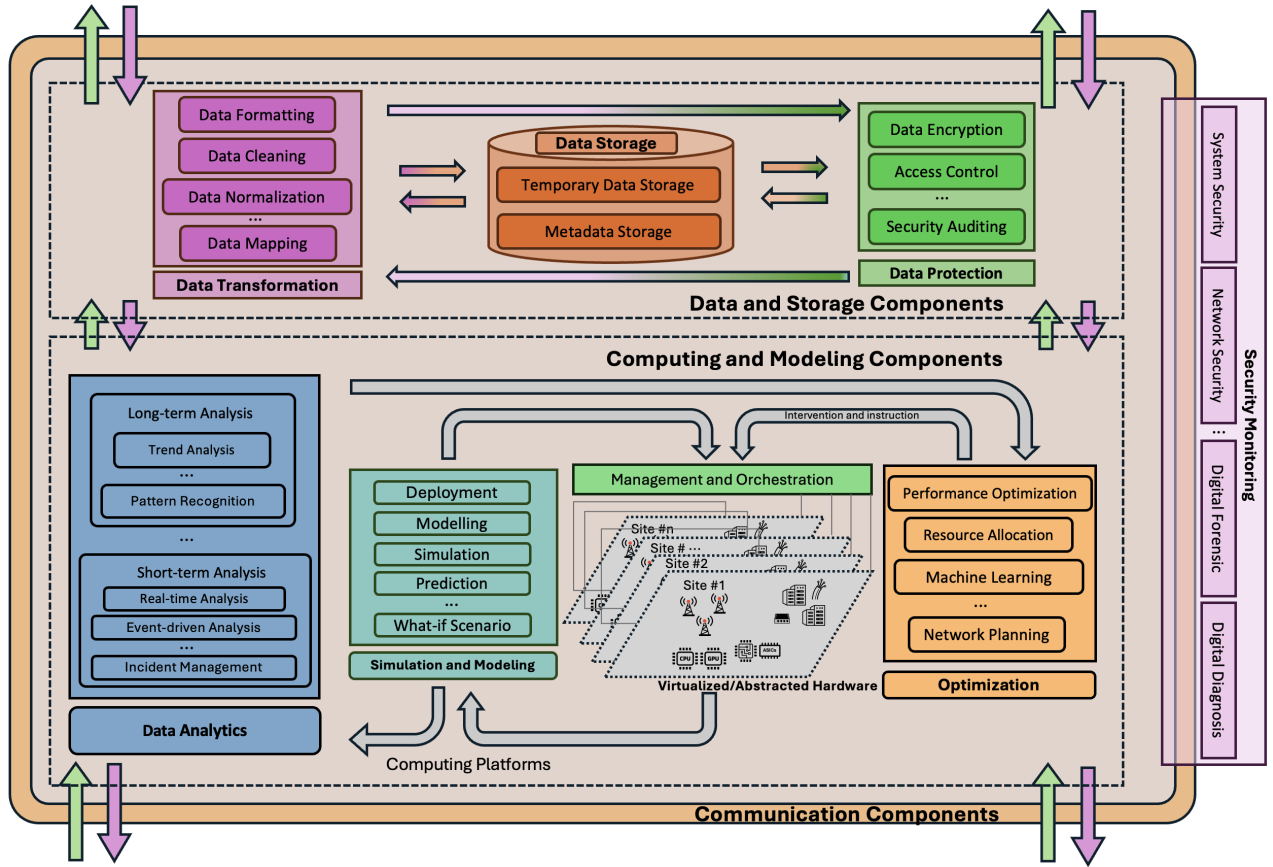


Figure 3: Digital Twin Layer, with Communication, Computing and Modeling and Data Storage Components

3.2.3 Computing and Modeling Components. In terms of composition, it encompasses two parts:

- **Virtualized Hardware:** This part involves the virtualized hardware, primarily displayed in the central part of Figure 3. A common method is to model or abstract the underlying hardware and environment.
- **Containerized software:** This part involves the implementation of service and network functions as software modules. The introduction of Software-defined Networking (SDN) enables flexible network management by separating the control and data planes, which allows more centralized control of the network.

In terms of functionality, the process starts with simulation and modeling. It models the network infrastructure and then initiates the process and logs by facilitating deployment, simulation, and what-if scenarios to evaluate network behavior. The data generated from this process is then handled by the data analytics block, which performs both long-term and short-term analysis, including real-time incident management and pattern recognition. The outcomes, in turn, can serve as inputs for deeper optimization and planning,

allowing interventions in the virtual network by network management and orchestration. In the optimization phase, AI models can enhance or replace traditional approaches for tasks such as resource allocation, congestion control, and network slicing [9]. This layer is designed with extensive scalability, allowing it to adapt and expand as network requirements grow, making it highly flexible for future developments and diverse use cases.

Additionally, these functions are integrated into a system that ensures smooth and secure operations. After thorough evaluation, instructions can be relayed to the physical network for real-world implementation and modification.

3.3 Application and Service Layer

As shown in Figure 4, the Application and Service Layer is the uppermost tier of the architecture, orchestrating services and enhancing user experiences. It incorporates applications like augmented reality, autonomous vehicles, smart cities, and telemedicine, designed to integrate with daily activities and infrastructure.

This layer communicates with the Digital Twin Layer and features a Service-oriented Architecture (SOA) for modular and reusable services. Modularity allows independent deployment to meet user demands without affecting the overall system. Furthermore, it supports isolated operations, enabling individual services to function in standalone mode, enhancing fault tolerance and reducing dependencies. This isolation is essential for ensuring that disruptions in one service do not cascade to others, thereby maintaining robustness and continuity. Each service should be resilient and tailored to evolving user needs.

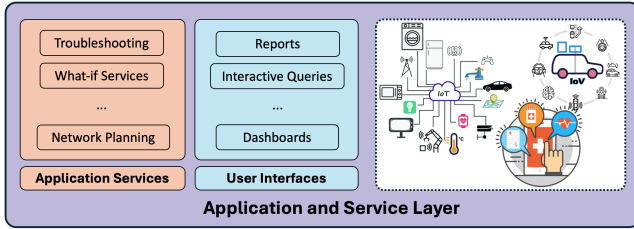


Figure 4: Application and Service Layer

4 Benchmarking Software Tools towards NDTs

In our 6G NDTs system architecture, the Digital Twin Layer plays the most important role, with the digital network as its core. Certain twin functions depend on simulation. For example, Ericsson's 5G digital twin uses simulation and reinforcement learning to train and adjust network parameters [5]. Therefore, it is essential to prioritize the choice of implementation tools for simulations.

A network simulator models behavior, while an emulator more accurately replicates it by simulating hardware and software conditions. Although simulators and emulators provide a solid foundation for an NDTs system, they must integrate real-time data, continuously updating the model and maintaining a balance between scalability, simulation fidelity, and performance.

The lack of a standardized 6G architecture requires tools to be extensible for future development. Simulators must accurately replicate 6G networks and be easily updated as standards evolve. Nevertheless, scalability and extensibility are challenging to quantify. Scalability and extensibility are challenging to quantify, but performance overhead on the network's common parts can be relatively compared through experiments. Therefore, experiments can be conducted to relatively compare performance overhead.

4.1 Resource Performance

Simulators and emulators have varying performance overheads. Tools like GNS3 use QEMU for hardware emulation, enabling more accurate network modeling by mimicking real hardware. Some other simulators focus solely on abstracting network behaviors, consuming fewer resources but capturing less complexity.

In order to briefly verify the differences between some potential tools, we conduct experiments to compare the resource utilization of different network simulators/emulators under the same topology deployments and functions in a containerized environment. We choose NS-3, GNS3, and Mininet, as they are non-commercial versions, which allows for customization, and each has its own unique characteristics, as well as representative examples from their respective categories. Among them, NS-3 is a discrete-event simulator in C++. Mininet and GNS3 are emulators developed by Python, but the former uses network namespace and virtual Ethernet interface to create topologies, and the latter uses virtualization technology such as QEMU, or Virtual Box to run the actual network operating system.

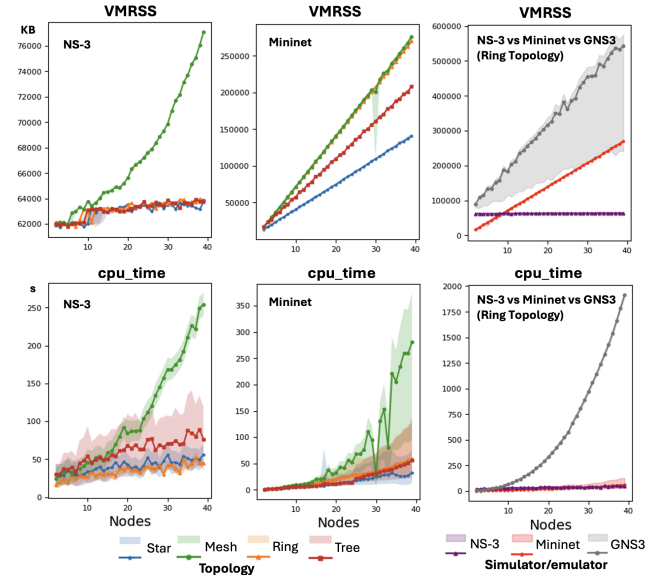


Figure 5: Simulators and emulators benchmark in different network sizes and topologies

We selected Virtual Memory Resident Set Size (VMRSS) in kilobytes (KB) to measure memory consumption, and CPU time in seconds (S) to measure CPU usage. The comparison focuses on how computing resources vary when handling dynamic changes in network scale and topology, with each node intermittently participating in UDP communication. For emulators, all associated and child processes are included.

Experiments show significant variation in resource usage among simulators/emulators in Figure 5. In general, as network complexity and scale increase, resource demands, especially CPU and memory, also grow, with demands accelerating as the network becomes more complex, for instance, fully connected mesh topology.

By conducting a cross-comparison, we can find that NS-3 has better results and more stable usage of resources, especially in memory. The CPU overhead of emulating can be several times that of simulating, and emulation also consumes significantly more memory. At the same time, emulators such as GNS3 also have higher overall usage due to emulating real IOS images, which provides higher scalability, even though the hardware requirements are the highest. Notably, these results are sensitive to specific setups, and small design differences can yield varying outcomes, though overall trends remain consistent.

4.2 Tools Selection

The concepts of Network Functions Virtualization (NFV) and SDN are applicable to both 5G and 6G, and give possible ideas and references for choosing appropriate network tools to build a full-stack 6G NDTs system. Since the Physical Twin Layer mainly consists of actual network infrastructure, and the Application and Service Layer is more like the interface exposed to higher layers, specific requirements can be addressed like the way of software development. Therefore, we should pay more attention to the Digital Twin Layer.

The main simulation challenges are in RAN and core networks. Open-source projects like free5GC, Open5GS, and UERANSIM support 5G networks, and extensions like 5G-LENA and ns3-mmwave leverage NS-3's extensibility. Combining free5GC with UERANSIM is a successful approach in 5G, and some modules can be reused in 6G, though the core network may need new tools.

For simulating physical environments, Unity3D, Unreal Engine, and Nvidia Omniverse are promising. Interfaces between the physical and digital worlds pose challenges, requiring efficient synchronization using tools like MQTT, Kafka, or RabbitMQ, or developing custom solutions. Data and Storage Components should retain time series data using appropriate databases and distributed storage, and graph databases can also be considered.

Like 5G, 6G should at least meet the key requirements mentioned above for fully deploying a twins system. The extreme KPI feature of 6G also adds additional complexity. Given that 6G NDTs are complex systems with numerous services and applications, and AI/ML interfaces should also be integrated in a distributed manner, it is essential to consider combining multiple tools, ensuring sufficient extensibility and well-defined interfaces.

5 Conclusion

In this paper, we have presented an architectural framework for implementing NDTs within the context of 6G networks. We proposed a three-layer architecture where each is designed to address the challenges of managing the next-generation mobile network environment. Through benchmarking different simulation and emulation tools, we showed the impact of tool selection on system performance, explaining the need for careful consideration when choosing technologies for NDTs system deployment.

As 6G technology continues to evolve, future work will focus on refining the proposed architecture, particularly in finding specialized interfaces for seamless synchronization between physical and digital layers. Additionally, we will explore integrating advanced AI-driven analytics and federated learning techniques to enhance the predictive capabilities and safety of 6G NDTs systems.

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